

YUKI OYAMA

PERSONAL DATA

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WORK EXPERIENCE

since 04/2024	Associate Professor Department of Civil Engineering, The University of Tokyo
04/2021–03/2024	Associate Professor Department of Civil Engineering, Shibaura Institute of Technology
04/2020–03/2021	Assistant Professor Department of Civil Engineering, Shibaura Institute of Technology
10/2019–03/2020	Project Assistant Professor Research Center for Advanced Science and Technology, The University of Tokyo
10/2017–09/2019	Research and Teaching Associate Transport and Mobility Laboratory (TRANSP-OR), École Polytechnique Fédérale de Lausanne (EPFL)
04/2017–09/2017	Research Fellow (PD) Japan Society for the Promotion of Science (JSPS), Tokyo Institute of Technology
04/2014–03/2017	Research Fellow (DC1) Japan Society for the Promotion of Science (JSPS), The University of Tokyo

VISITING RESEARCHER

since 04/2022	Visiting Associate Professor Ehime University, Urban Design Center Matsuyama
since 04/2020	Senior Visiting Researcher Research Center for Advanced Science and Technology, The University of Tokyo
11/2016–03/2017	Visiting Scholar Transport and Mobility Laboratory (TRANSP-OR), École Polytechnique Fédérale de Lausanne (EPFL)
05/2014–03/2015	Visiting Researcher Ehime University, Urban Design Center Matsuyama

SCIENTIFIC EDUCATION

03/2017	PH.D IN ENGINEERING Behavior in Networks Studies Unit (Supervisor: Eiji Hato) Department of Urban Engineering, The University of Tokyo
03/2014	M.SC. IN ENGINEERING Behavior in Networks Studies Unit (Supervisor: Eiji Hato) Department of Urban Engineering, The University of Tokyo
03/2012	B.SC. IN ENGINEERING Environmental Design Laboratory (Supervisor: Mikiko Ishikawa) Department of Urban Engineering, The University of Tokyo

PEER-REVIEWED JOURNAL PAPERS

(*: Corresponding author; # Students supervised by me.)

International Journal Papers

1. Huang, H.K.#, Okazaki, R.#, **Oyama, Y.*** (2026) A practical approach to multi-day dynamic pricing in attended home delivery. *Transportation Research Part E: Logistics and Transportation Review* 215: 105078.
2. Imamura, K.#, **Oyama, Y.*** (2026) Optimizing loading space locations for walkable city centers. *Transportation Research Part A: Policy and Practice* 211: 105112.
3. Wang, J., **Oyama, Y.*** (2026) Quantifying willingness-to-walk across varying urban visual contexts using a global-local path choice model. *Journal of Transport Geography* 130: 104502.
4. Gramsch-Calvo, B.*, Okamura, K., Takami, K., **Oyama, Y.**, Chikaraishi, M., Axhausen, K.W., Parady, G. (2025) Going the extra mile: Estimating the willingness to travel to meet with friends using a joint destination choice model. *Journal of Transport Geography* 129: 104429.
5. Okazaki, R.#, **Oyama, Y.***, Imura, N., Nishinari, K. (2025) Evaluating choice-based demand management strategies for day-to-day home delivery planning. *Research in Transportation Economics* 113: 101615.
6. Parady, G.*, Chikaraishi, M., **Oyama, Y.** (2025) A walker's paradise ain't a driver's hell: Evaluating the causal effect of temporary road pedestrianization on traffic conditions of surrounding roads. *Journal of Transport Geography* 127: 104269.
7. **Oyama, Y.***, Akamatsu, T.* (2025) A market-based efficient matching mechanism for crowd-sourced delivery systems with demand/supply elasticities. *Transportation Research Part C: Emerging Technologies* 174: 105110.
8. Akamatsu, T.*, **Oyama, Y.*** (2024) A fluid-particle decomposition approach to matching market design for crowdsourced delivery systems. *Transportation Research Part C: Emerging Technologies* 166: 104738.
9. **Oyama, Y.***, Murakami, D., Krueger, R. (2024) A hierarchical Bayesian logit model for spatial multivariate choice data. *Journal of Choice Modelling* 52: 100503.
10. **Oyama, Y.***, Murakami, S., Chikaraishi, M., Parady, G. (2024) Designing pedestrian zones within city center networks considering policy objective trade-offs. *Transportation Research Part A: Policy and Practice* 185: 104119.
11. **Oyama, Y.*** (2024) Spatial city image and its formative factors: A street-based neighborhood cognition analysis. *Cities* 149: 104898.
12. **Oyama, Y.*** (2024) Global path preference and local response: A reward decomposition approach for network path choice analysis in the presence of visually perceived attributes. *Transportation Research Part A: Policy and Practice* 181: 103998.
13. **Oyama, Y.***, Fukuda, D., Imura, N., Nishinari, K. (2024) Do people really want fast and precisely scheduled delivery? E-commerce customers' valuations of home delivery timing. *Journal of Retailing and Consumer Services* 78: 103711.
14. Parady, G.*, **Oyama, Y.**, Chikaraishi, M. (2023) Text-aided Group Decision-making Process Observation Method (x-GDP): A novel methodology for observing the joint decision-making process of travel choices. *Transportation*.
15. **Oyama, Y.*** (2023) Capturing positive network attributes during the estimation of recursive logit models: A prism-based approach. *Transportation Research Part C: Emerging Technologies* 147: 104014.
16. Parady, G.*, Suzuki, K., **Oyama, Y.**, Chikaraishi, M. (2023) Activity detection with Google Maps Location History data: factors affecting joint activity detection probability and its potential application on real social networks. *Travel Behaviour and Society* 30: 344-357.
17. **Oyama, Y.***, Hara, Y.*, Akamatsu, T.* (2022) Markovian traffic equilibrium assignment based on network generalized extreme value model. *Transportation Research Part B: Methodological* 155: 135-159.
18. **Oyama, Y.***, Hato, E. (2019) Prism-based path set restriction for solving Markovian traffic assignment problem. *Transportation Research Part B: Methodological* 122: 528-546.

19. Oyama, Y.^{*}, Hato, E. (2018) Link-based measurement model to estimate route choice parameters in urban pedestrian networks. *Transportation Research Part C: Emerging Technologies* 93: 62-78.
20. Oyama, Y.^{*}, Hato, E. (2017) A discounted recursive logit model for dynamic gridlock network analysis. *Transportation Research Part C: Emerging Technologies* 85: 509-527.

National Journal Papers (in Japanese)

21. Kizaki, R., Yaginuma, H.^{*}, Oyama, Y., Terabe, S., Suzuki, Y. (2024) Dynamic pedestrian route choice model considering street landscape and spatial characteristics. *Journal of Japan Society of Civil Engineers D3 (Infrastructure Planning and Management)* 80(20): 24-20064.
22. Okamura K., Oyama, Y., Chikaraishi, M., Takami, K., Parady, G.^{*} (2024) Modeling collective decisions on eating-out location choice. *Journal of the City Planning Institute of Japan* 59(3): 1675-1682 (in Japanese).
23. Tojo, T.[#], Oyama, Y.^{*} (2022) A deep learning model for building type estimation based on building names. *Journal of the City Planning Institute of Japan* 57(3): 1025-1032. (in Japanese)
24. Matsumura, K.[#], Oyama, Y.^{*} (2022) Efficiency analysis of a capacitated MaaS system focusing on different payment schemes. *Journal of the City Planning Institute of Japan* 57(3): 666-673. (in Japanese)
25. Murakami, S.[#], Oyama, Y.^{*} (2022) Optimal location of pedestrian streets based on a multimodal equilibrium assignment model. *Journal of the City Planning Institute of Japan* 57(3): 622-629. (in Japanese)
26. Ikegami, T.[#], Oyama, Y.^{*} (2021) Proposal of an activity-based elderly mobility index focusing on public transportation convenience. *Journal of the City Planning Institute of Japan* 56(3): 563-570. (in Japanese)
27. Oyama, Y.^{*}, Hato, E. (2017) Structural estimation for route choice models considering link specificity of measurement error variances. *Journal of Japan Society of Civil Engineers, Ser. D3 (Infrastructure Planning and Management)* 73(5): I_597-I_608 (in Japanese).
28. Oyama, Y.^{*}, Hato, E. (2017) Stochastic assignment in time-structured networks. *Journal of Japan Society of Civil Engineers, Ser. D3 (Infrastructure Planning and Management)* 73(4): 186-200 (in Japanese).
29. Oyama, Y.^{*}, Hato, E. (2017) Pedestrian activity-based network design based on multi-objective programming. *Journal of the City Planning Institute of Japan* 52(3): 810-817 (in Japanese).
30. Oyama, Y., Hato, E.^{*} (2017) Route choice analysis in a disaster network using generalized recursive logit model. *JSTE Journal of Traffic Engineering* 3(5): 1-10 (in Japanese).
31. Oyama, Y.^{*}, Hato, E. (2016) Pedestrian activity assignment problem with time-space constraint and path correlation. *Journal of the City Planning Institute of Japan* 51(3): 680-687 (in Japanese).
32. Oyama, Y.^{*}, Fukuyama, S., Hato, E. (2014) A study on mechanisms of short trip generations using discrete-continuous model based on activity-needs. *Journal of the City Planning Institute of Japan* 49(3): 375-380 (in Japanese).
33. Oyama, Y., Hato, E.^{*} (2013) Move-stay choice model using Voronoi-based overlay networks. *Journal of the City Planning Institute of Japan* 48(3): 1107-1112 (in Japanese).
34. Oyama, Y., Hato, E.^{*} (2012) Route choice model based on continuity of streetscapes. *Journal of the City Planning Institute of Japan* 47(3): 643-648 (in Japanese).

PEER-REVIEWED INTERNATIONAL CONFERENCES

1. Wang, J., Oyama, Y. (2025) A Markovian Parking Search Equilibrium model with an application to optimization of curbside logistics loading spaces. The 29th International Conference of Hong Kong for Transportation Studies (HKSTS).
2. Kadosaka, Y.[#], Oyama, Y., Le, Y. (2025) Modeling destination search behavior considering cognitive process of dynamically updated information. The 29th International Conference of Hong Kong for Transportation Studies (HKSTS).
3. Huang, H.K.[#], Okazaki, R.[#], Oyama, Y. (2025) Balancing Efficiency and Service: A Marginal Cost Approximation Approach to Multi-Period Vehicle Routing in E-Commerce Logistics. The IEEE International Conference on Intelligent Transportation Systems (ITSC) 2025.

4. Okazaki, R.[#], **Oyama, Y.** (2025) User-based time slot pricing for day-to-day home delivery with routing cost approximations. Transportation Research Board Annual Meeting 2025.
5. Imamura, K.[#], **Oyama, Y.** (2025) Optimal loading space locations for walkable cities. Transportation Research Board Annual Meeting 2025.
6. Gramsch-Calvo, B., Okamura, K., Takami, K., **Oyama, Y.**, Chikaraishi, M., Axhausen, K.W., Parady, G. (2025) Going the extra mile: Estimating the willingness to travel to meet with friends using a joint destination choice model. Transportation Research Board Annual Meeting 2025.
7. **Oyama, Y.** (2024) Pedestrians' local responses to visual streetscape quality: A comparative analysis of walking behavior across different urban contexts. Urban Transitions 2024, Sitges, Spain.
8. Parady, G., Chikaraishi, M., **Oyama, Y.** (2024) Optimization-enriched Interactive Street Network Design (i-SND): A novel methodology for human-centric urban transportation planning. Urban Transitions 2024, Sitges, Spain. [\[Poster\]](#)
9. Fujiwara, K., **Oyama, Y.**, Chikaraishi, M., Fujiwara, A. (2024) Pedestrian dynamics and their policy implications for urban public space design. Urban Transitions 2024, Sitges, Spain.
10. Kizaki, R., **Oyama, Y.**, Terabe, S., Suzuki, Y., Yaginuma, H. (2024) Evaluation of street space development based on a pedestrian route choice model considering street landscape. The 17th International Conference on Travel Behaviour research (IATBR), Wien, Austria.
11. Safitri, N.D., **Oyama, Y.**, Chikaraishi, M. (2024) How random is route choice behavior during disaster? Inverse estimation of perceived travel time using a recursive logit model with structured variance. The 17th International Conference on Travel Behaviour research (IATBR), Wien, Austria.
12. Fujiwara, K., **Oyama, Y.**, Chikaraishi, M., Fujiwara, A. (2024) Pedestrian behavior modeling representing competitive nature between movers and stayers in urban public space: A numerical simulation. The 17th International Conference on Travel Behaviour Research (IATBR), Wien, Austria.
13. Parady, G., **Oyama, Y.**, Chikaraishi, M. (2024) Text-aided modelling of group decision-making processes: An application to eating-out destination choice. The 17th International Conference on Travel Behaviour Research (IATBR), Wien, Austria.
14. **Oyama, Y.** (2024) A link-based route choice model with decomposed reward functions: An application to walkability analysis. The 17th International Conference on Travel Behaviour research (IATBR), Wien, Austria.
15. **Oyama, Y.**, Murakami, S.[#], Parady, G., Chikaraishi, M. (2024) Designing Pedestrian Zones within City Center Networks Considering Policy Objective Trade-offs. Transportation Research Board (TRB) Annual Meeting 2024, Washington D.C., US.
16. Parady, G., Chikaraishi, M., **Oyama, Y.** (2024) A Walker's Paradise ain't a Driver's Hell: Evaluating the Causal Effect of Temporary Road Pedestrianization on Traffic Conditions of Surrounding Roads. Transportation Research Board (TRB) Annual Meeting 2024, Washington D.C., US.
17. Okazaki, R.[#], **Oyama, Y.**, Imura, N., Nishinari, K. (2023) Day-to-day delivery demand management: Evaluation based on routing efficiency and customer satisfaction. The 11th symposium arranged by European Association for Research in Transportation (hEART), Zurich, Switzerland.
18. Parady, G., **Oyama, Y.**, Chikaraishi, M. (2023) Text-aided Group Decision-making Process Observation Method (x-GDP): A novel methodology for observing the joint decision-making process of travel choices. The 11th symposium arranged by European Association for Research in Transportation (hEART), Zurich, Switzerland.
19. Fujiwara, K., **Oyama, Y.**, Chikaraishi, M., Fujiwara, A. (2023) A pedestrian model capturing trade-offs between travel and place functions of urban street space: An illustrative numerical example. The 15th International Conference of the Eastern Asia Society for Transportation Studies (EASTS 2023), Shah Alam, Malaysia. [\[Proceedings\]](#)
20. Okazaki, R.[#], **Oyama, Y.**, Imura, N., Nishinari, K. (2023) Evaluation of choice-based demand management strategies for day-to-day delivery planning. The 5th Bridging Transport Researchers (BTR5) conference, online.
21. Murakami, S.[#], **Oyama, Y.** (2023) Multimodal network design for the optimal placement of pedestrian streets in an urban congested network. The 18th International Conference on Computational Urban Planning and Urban Management (CUPUM), Montreal, Canada.

22. **Oyama, Y.** (2022) A prism-constrained recursive logit model to analyze positive utilities in pedestrian route choice behavior. The 16th International Conference on Travel Behaviour research (IATBR), Santiago, Chile.
23. Parady, G., **Oyama, Y.**, Chikaraishi, M. (2022) Understanding the joint decision-making process of leisure destination choices: Exploring new methodologies. The 16th International Conference on Travel Behaviour research (IATBR), Santiago, Chile. [\[Presentation\]](#)
24. Suzuki, K., Oyama, Y., Chikaraishi, M., Parady, G. (2021) The effectiveness of using Google Maps Location History data to detect joint activities in social networks. The 3rd Bridging Transportation Researchers (BTR) Conference, Online. [\[Presentation\]](#)
25. **Oyama, Y.**, Scarinci, R., Bierlaire, M. (2018) Optimal capacity location problem of parking and accelerated moving walkways to design a car-free city center. The 7th symposium arranged by European Association for Research in Transportation (hEART), Athens, Greece.
26. **Oyama, Y.**, Hato, E., Scarinci, R., Bierlaire, M. (2017) Markov assignment for a pedestrian activity-based network design problem. The 6th symposium arranged by European Association for Research in Transportation (hEART), Haifa, Israel. [\[Presentation\]](#)
27. **Oyama, Y.**, Hato, E. (2016) Pedestrian activity model based on implicit path enumeration. Proceedings of the 21st International Conference of Hong Kong for Transportation Studies (HKSTS), pp.331-338.
28. **Oyama, Y.**, Hato, E. (2016) A link-based map matching algorithm with structural estimation method. The 5th symposium arranged by European Association for Research in Transportation (hEART), Delft, Netherland.
29. **Oyama, Y.**, Hato, E. (2016) Incorporating destination choices into microscopic land-use dynamics modeling using longitudinal data. The 9th Triennial Symposium on Transportation Analysis (TRISTAN), Oranjestad, Aruba.
30. **Oyama, Y.**, Chikamatsu, K., Shoji, Y., Hato, E., Koga, M. (2016) Trajectory-oriented traffic management using sequential discount rate: a case study of the Great East Japan Earthquake. The 11th ITS European Congress (ERTICO), Glasgow, Scotland.
31. **Oyama, Y.**, Hato, E. (2015) Incorporating context-dependent energy into the pedestrian dynamic scheduling model with GPS data. The 14th International Conference on Travel Behaviour research (IATBR), Windsor, England. [\[Presentation\]](#)
32. Kokubun, A., Hato, E., **Oyama, Y.** (2013) Modeling Transformation in Neighborhood with Urban Redevelopment in Built-up Area of Tokyo Incorporating Effect of Interaction. Proceedings of International Conference on Computers in Urban Planning and Urban Management (CUPUM), Utrecht, Netherland.
33. **Oyama, Y.**, Hato, E. (2013) Mobility Design: Incorporating "Private" Public Space onto the Street for the City Conversion. Proceedings of 5th International Congress of International Association of Societies of Design Research (IASDR), pp.3235-3244.

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